

Tools for Administering an Oracle Database:

⇒ following tools are used for installation and upgrade.

(i) Oracle Universal Installer (OUI):

- Installs, ^{upgrade and uninstalling the} Oracle software and options.
- Can automatically launch the database.

Configuration assistant to create a database.

(ii) Database Configuration Assistant (DBCA): ^{Provides GUI. Guides user through installation process.}

- Creates a database from Oracle-supplied templates, enabling to copy a preconfigured seed database.

Automates many common tasks such as creating DB instances, configuring parameters and setting up options.

(iii) Database Upgrade Assistant (DBUA):

- Guides through the upgrade of existing database to a new Oracle release.

Facilitates upgrading databases to newer versions. Automates many tasks involved in upgrading such as database backup, schema changes and parameter adjustment.

(iv) Oracle Net Manager:

- Configures network connectivity for Oracle databases and applications.

- Part of Oracle networking software.

- Enables setup of databases connections.

- Listener and naming methods.

- Helps ensure efficient communication between Oracle clients and servers.

The following tools are used to manage Oracle instance and database:

OEM:

◦ Oracle Enterprise Manager is a management toolset provided by Oracle Corporation for managing and monitoring Oracle's software stack, including databases, middleware, and applications, all from

a single interface.

⇒ It is a system management tool which provides an integrated solution for managing heterogeneous environment.

⇒ It combines a graphical console, agents, common services and tools to provide an integrated, comprehensive systems management platform for managing Oracle products.

From the Oracle Enterprise Manager Console, the following tasks can be performed:

⇒ Administer, diagnose, and tune multiple databases.

⇒ Distribute software to multiple servers and clients.

⇒ Schedule jobs on multiple nodes at varying time intervals.

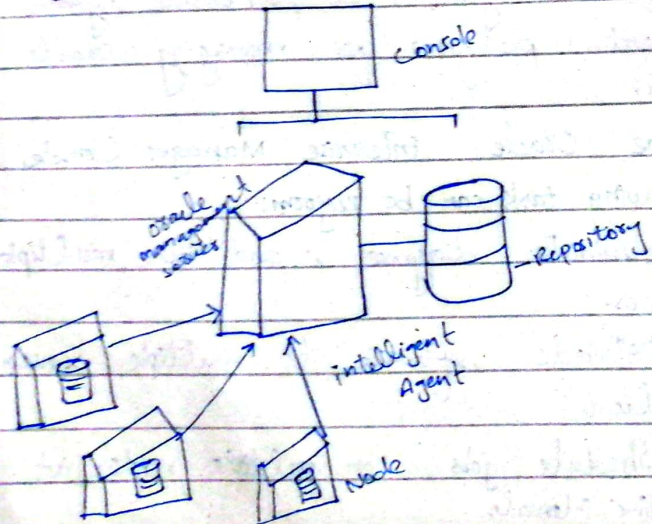
⇒ Distribute software to multiple servers and clients.

⇒ Monitor objects and events throughout the network.

⇒ Administer Real Application Clusters

⇒ Customize display using multiple graphic maps and groups of network objects, such as nodes and databases.

⇒ Overall OEM serves as a comprehensive solution for managing and monitoring Oracle environments, offering a centralized platform for administration, automation and optimization of Oracle products and services.



⇒ The main EM tools that are used to administer an Oracle database are:

Enterprise Manager Database Console

◦ To administer one database.

Enterprise Manager Grid Control

◦ To administer many databases at the same time.

Recovery Manager

⇒ Oracle tool that provides a complete solution for the backup, restoration, and recovery needs of the entire database or of specific database files.

Oracle Secure Backup

⇒ Provides tape backup management for the Oracle ecosystem, which includes:

- Oracle database protection to tape through integration with Recovery Manager

◦ Seamless Support of Oracle Real Application Clusters

- Central administration of distributed clients and media servers, Oracle Collaboration Suits, Oracle home and binaries.

Data Pump

- Oracle data pump is a feature used for fast and efficient data movement and transformation in Oracle databases.
- moves large volumes of data between Oracle database quickly & efficiently.
- Allows exporting data from a database to a dump file and importing it into another database.
- Transfers both table data and metadata, including definitions, constraints and indexes.
- Can transfer data directly on network without intermediate storage.

SQL *Loader

- Enables the loading of data from an external file into an

Oracle database

⇒ Efficiently loads large volumes of data into Oracle tables from external files.

⇒ Requires a control file that specifies how the data in the external file should be loaded into the tables of database.

⇒ The control file defines the format of the data, mapping between columns in the file and columns in the table and any data transformation rules.

⇒ Supports various data formats such as delimited files, fixed-width files and other custom formats.

⇒ Have robust error handling capabilities.

Command-line tools

To administer enterprise managers

emctl start | status | set | stop dbconsole

To administer the listener:

lsnrctl help | start | status | stop
dbconsole

Installation: System requirements

Memory requirements

- o 1-GB for the instance with Database Control.

Disk space requirements

- o 1.5 GB of swap space
- o 400MB of disk space in the tmp directory
- o Between 1.5 GB and 3.5 GB for the Oracle Software.
- o 1.2 GB for the preconfigured database
- o 2.4 GB for the flash recovery area.
- o OS
 - o Adequate temporary space
 - o 64-bit versus 32-bit issues
 - o OS patch level
 - o System packages
 - o System and kernel parameters
 - o X server permissions
 - o Sufficient swapping
 - o Nonempty ORACLE_HOME